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Preface

Human interaction is often described through devices, technologies, or cultural practices.

But beneath these layers lies a deeper structure—one that is natural, universal, and enduring.

The NAW Human Interface Framework (NAW HIF) was born from a simple question:

“What is the most durable way to describe how humans perceive, interpret, and act?”

This document is the result of years of structural refinement, philosophical distillation, and terminological precision.

It offers a minimal yet complete system—one that can be used by educators, designers, analysts, and learners alike.

NAW HIF is not a theory.

It is a framework: a reference system that allows us to see clearly, act coherently, and design responsibly.

It is built to last—not just for today’s technologies, but for the next hundred years of human–machine interaction.

I invite you to read slowly, reflect deeply, and apply freely.

The structure is yours to use.

— S. Hara

Overview Summary

The NAW Human Interface Framework (NAW HIF) is a structural model that defines how humans perceive, interpret, and act within any environment.

It reconstructs human interfaces as a natural structure—a system that does not depend on specific technologies, devices, or cultural conditions, and can therefore remain valid across long periods of time.

At its core, NAW HIF organizes human interaction into a three-layer model:

- Primary Interface Domain (6) — the fundamental sensory and internal channels through which information enters the human system.
- Contextual Interface Domain (4) — the situational modes that determine how information is received and processed.
- Interaction Device Domain (2) — the means through which humans act upon or interact with the external world.

Together, these domains form a 6-4-2 structure that captures the full cycle of perception, cognition, and action.

This structure provides a stable reference frame that allows individuals to recognize which layer and which context their current behavior or learning belongs to.

NAW HIF is designed not only to describe human interaction, but also to serve as a technical system for understanding and improving behavior, learning, and interface design.

By defining clear layers, domains, and contexts, the framework enables:

- High-speed and reproducible understanding of human behavior
- Cross-domain analysis through consistent structural units
- Prevention-oriented design, reducing problems before they arise
- Long-term durability, ensuring the framework remains usable for decades

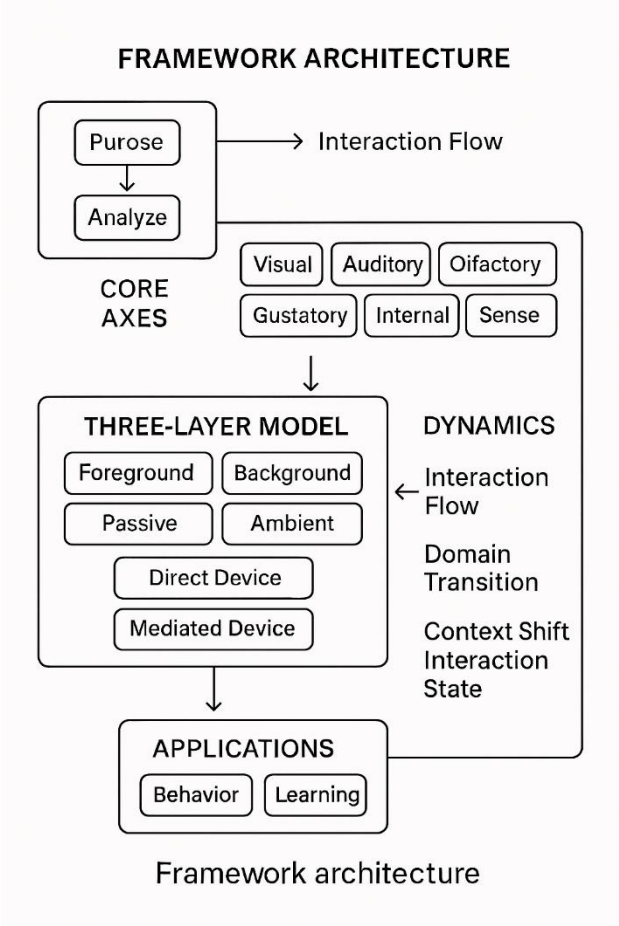
The framework is guided by five core principles—Architecture, Orientation, Reproducibility, Prevention, and Durability—and is further supported by design values such as continuity with traditional human practices, harmony with existing concepts, ease of learning, elimination of redundancy, controlled information density, and long-term usability.

NAW HIF is positioned as the foundational layer of the broader NAW system, connecting naturally to the Natural Brain (NDB)

series, which addresses human intelligence and information management.

Together, these works aim to establish a unified, future-proof standard for understanding human cognition and interaction in an age where digital and natural systems increasingly converge.

Figure 1. NAW Human Interface Framework Architecture Overview



Terminology

A. Organization

NAW (Natural Architecture Works)

A project that reconstructs human intelligence, behavior, and learning as a “natural structure” that does not depend on specific technologies or environments, designed for long-term stability and universal applicability.

NIL (Natural Intelligence Lab)

The conceptual research laboratory responsible for developing, validating, and advancing the NAW framework and its associated systems.

B. Framework – Core Structure

NAW Human Interface Framework (NAW HIF)

A foundational framework that defines human interfaces through a three-layer structure composed of six Primary interfaces, four Contextual interfaces, and two Device interfaces.

Framework Architecture

The structural organization of NAW HIF, describing how layers, domains, contexts, and devices interrelate to form a coherent system.

Three-Layer Model

The core structural model of NAW HIF consisting of the Primary Interface Domain, the Contextual Interface Domain, and the Interaction Device Domain.

Natural Structure

A structure derived from inherent human perception, cognition, and behavior, independent of technological conditions or environmental constraints.

Human Interface

The natural entry points through which humans perceive the world, process information, and generate action.

Layer

A structural level indicating where processing, perception, or interaction occurs within the framework.

Domain

A categorical unit within the framework, consisting of the Primary, Contextual, and Device domains.

Interface

A point of contact or connection between the human system and the external world.

Mode

A state or operational pattern that defines how interaction is carried out.

Context

The situational environment in which information is received, interpreted, and acted upon.

System

An integrated structure in which multiple layers, domains, and contexts function together coherently.

Reference Frame

A structural basis for recognizing one's position, state, and context within the framework.

Stability

The property of maintaining consistent function, meaning, and usability over time.

Universality

The property of being applicable across cultures, technologies, and historical periods.

C. Framework – Domains and Interfaces (6-4-2)

1. Primary Interface Domain (6)

Visual

Reception of information through sight, including shape, motion, color, and spatial patterns.

Auditory

Reception of information through sound, including tone, rhythm, and verbal or non-verbal cues.

Tactile

Reception of information through physical contact, pressure, texture, and temperature.

Olfactory

Reception of information through smell, enabling detection of environmental and chemical cues.

Gustatory

Reception of information through taste, supporting evaluation of substances and internal states.

Internal Sense

Perception of internal bodily and cognitive states, including balance, proprioception, tension, breathing, emotion, and mental orientation.

2. Contextual Interface Domain (4)

Foreground Interaction

Active, intentional engagement with a task or stimulus.

Background Interaction

Parallel or peripheral engagement occurring while attention is directed elsewhere.

Passive Interaction

Reception of information without deliberate attention or effort.

Ambient Interaction

Continuous perception of environmental information as part of the surrounding context.

3. Interaction Device Domain (2)

Direct Device

A tool or medium operated through direct physical or cognitive contact (e.g., hands, voice, paper, body movement).

Mediated Device

A tool or medium requiring technological or physical mediation (e.g., screens, controllers, digital interfaces).

D. Framework – Design Principles

Architecture

The structural organization of human interfaces and intelligence as a natural system.

Orientation

The ability to recognize one's current layer, domain, and contextual position.

Reproducibility

The capacity for behaviors, skills, and learning processes to be replicated consistently and efficiently.

Prevention

A design philosophy that emphasizes avoiding problems structurally rather than reacting after they occur.

Durability

Long-term viability of definitions, structures, and concepts across generations and technological shifts.

E. Cross-Domain Dynamics

Cross-Domain Dynamics

The integrated behavior that emerges when Primary, Contextual, and Device domains interact as a unified system.

Interaction Flow

The movement of information across multiple domains during perception or action.

Domain Transition

Shifts between interface domains as perception or behavior changes.

Context Shift

Changes in situational or attentional context.

Processing Pathway

The route through which information travels across layers and domains.

Interaction State

A temporary configuration indicating which domain, layer, and context are currently active.

F. Application Examples

Use Case

A concrete example illustrating a specific application scenario.

Scenario

A narrative sequence describing how interaction unfolds over time.

Pattern

A recurring structural configuration or behavioral form.

Mapping

The act of aligning concepts, structures, or processes across layers or domains.

G. Category Keywords

UI/UX

User interface and user experience.

Cognitive Load

The mental resources required for information processing.

Human-Computer Interaction (HCI)

The study of interaction between humans and computational systems.

Digital Wellbeing

The maintenance of health and balance in digital environments.

H. Publications (Bridge Keywords)

Natural Brain I

A framework defining human intelligence as a “Natural Data Base.”

NDB (Natural Data Base System)

The core structural model introduced in Natural Brain I.

Natural Brain II

A framework addressing comprehensive human information management.

CIM (Comprehensive Information Management System)

The core structural model introduced in Natural Brain II.

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Chapter 1. Framework Architecture

1.1 Purpose of the Framework

The NAW Human Interface Framework (NAW HIF) defines human perception, interpretation, and action as a natural structure—a structure that does not depend on specific technologies, devices, or cultural environments.

Its purpose is to provide a stable and universal reference system that enables individuals and designers to understand which layer, which domain, and which context their current behavior or learning belongs to.

The framework aims to:

- Reconstruct human interfaces and intelligence as a coherent natural system
- Provide a structural reference that remains valid across generations
- Enable high-speed and reproducible understanding of human behavior
- Establish preventive standards that reduce problems before they arise
- Offer a long-term durable model suitable for education, design, and cognitive analysis

NAW HIF serves as the foundational layer of the broader NAW system, connecting naturally to the Natural Brain (NDB) series, which addresses human intelligence and information management.

1.2 Core Axes of NAW

The architecture of NAW HIF is guided by five core axes.

These axes define the orientation and long-term stability of the framework.

Architecture

The structural organization of human interfaces and intelligence as a natural system.

Orientation

The ability to recognize one's current layer, domain, and contextual position.

Reproducibility

The capacity for behaviors, skills, and learning processes to be replicated consistently and efficiently.

Prevention

A design philosophy that emphasizes avoiding problems structurally rather than reacting after they occur.

Durability

Long-term viability of definitions, structures, and concepts across generations and technological shifts.

These axes ensure that NAW HIF remains coherent, scalable, and applicable across diverse environments.

1.3 Design Principles

The framework is further supported by six design principles that ensure clarity, usability, and long-term relevance.

Continuity with Traditional Values

Respecting long-standing human practices and natural learning patterns.

Harmony with Existing Concepts

Integrating smoothly with established academic and practical frameworks.

Ease of Learning

Minimizing cognitive load through simple and intuitive structures.

Elimination of Redundancy

Removing overlapping or unnecessary concepts to maintain clarity.

Controlled Information Density

Presenting only essential information to support efficient understanding.

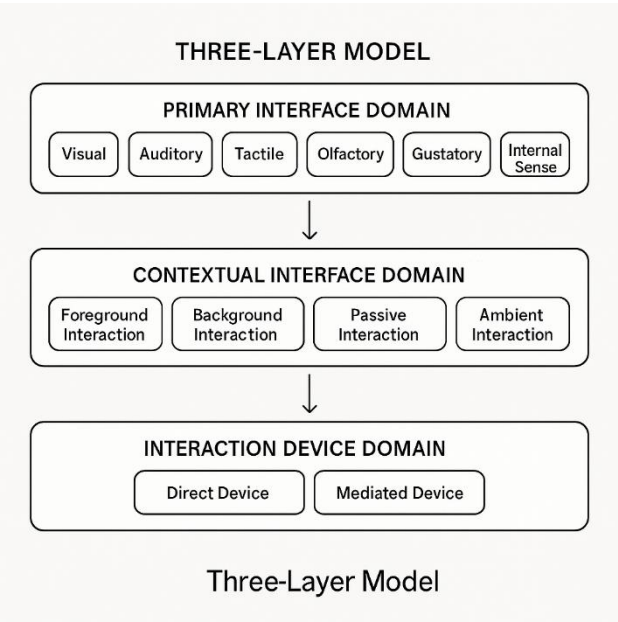
Long-Term Usability

Ensuring definitions and structures remain valid over extended periods.

These principles guide the formulation of terminology, diagrams, and examples throughout the framework.

1.4 Three-Layer Model

Figure 2. Three-Layer Model of Human Interaction



At the core of NAW HIF is the Three-Layer Model, which organizes human interaction into three integrated domains:

1. Primary Interface Domain (6)

The fundamental sensory and internal channels through which information enters the human system.

2. Contextual Interface Domain (4)

The situational modes that determine how information is received and processed.

3. Interaction Device Domain (2)

The means through which humans act upon or interact with the external world.

This 6-4-2 structure provides a complete and non-redundant description of perception, cognition, and action.

It functions as a stable reference frame that allows individuals to identify the structural position of any interaction.

1.5 Structural Components

Layer

A structural level indicating where processing, perception, or interaction occurs.

Domain

A categorical unit within the framework, consisting of the Primary, Contextual, and Device domains.

Interface

A point of contact or connection between the human system and the external world.

Mode

A state or operational pattern that defines how interaction is carried out.

Context

The situational environment in which information is received, interpreted, and acted upon.

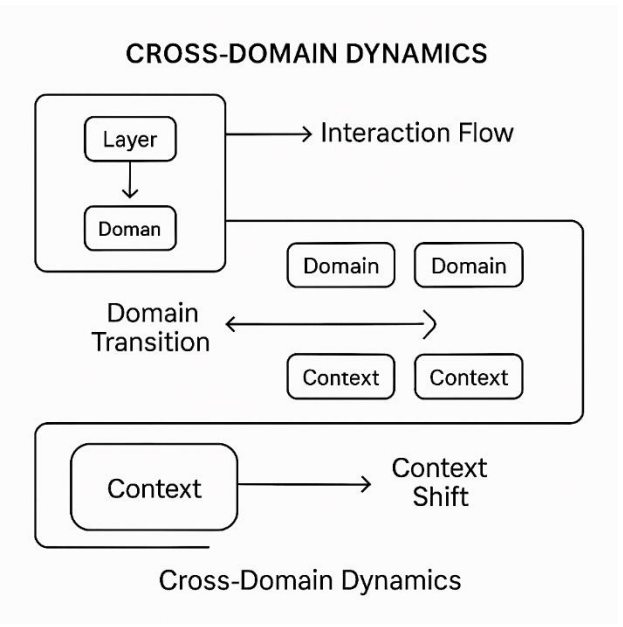
System

An integrated structure in which multiple layers, domains, and contexts function together coherently.

These components allow NAW HIF to describe human interaction with precision and consistency.

1.6 Cross-Domain Integration

Figure 3. Cross-Domain Dynamics and Interaction Flow



Human interaction rarely occurs within a single domain. Instead, perception, context, and device layers operate simultaneously.

NAW HIF captures this through the concept of Cross-Domain Dynamics, which includes:

- Interaction Flow

The movement of information across multiple domains.

- Domain Transition

Shifts between interface domains as perception or behavior changes.

- Context Shift

Changes in situational or attentional context.

- Processing Pathway

The route through which information travels across layers and domains.

- Interaction State

A temporary configuration indicating which domain, layer, and context are currently active.

These dynamics allow NAW HIF to describe real-time human activity with structural clarity.

1.7 Position within the NAW System

NAW HIF is the foundational layer of the broader NAW architecture.

It provides the interface-level structure upon which the Natural Brain (NDB) series is built:

- Natural Brain I (NDB)

Defines human intelligence as a Natural Data Base.

- Natural Brain II (CIM)

Defines comprehensive human information management.

Together, these works form a unified system for understanding human cognition, behavior, and interaction in a future where natural and digital systems coexist.

1.8 Scope and Intended Use

NAW HIF is designed for:

- Cognitive and behavioral analysis
- Educational design and learning systems
- UI/UX and interaction design
- Human-computer interaction research
- Personal development and self-orientation
- Long-term structural reference in digital and natural environments

The framework is intentionally minimal, universal, and durable, enabling broad application without loss of clarity.

1.9 Summary

Chapter 1 establishes the structural foundation of NAW HIF.

It defines the purpose, core axes, design principles, and the Three-Layer Model that organizes all subsequent chapters.

This architecture ensures that the framework remains coherent, reproducible, and applicable across diverse contexts, forming a stable foundation for the detailed domain definitions that follow

Chapter 2. Primary Interface Domain

2.1 Purpose of the Primary Interface Domain

The Primary Interface Domain defines the six fundamental channels through which humans receive information from the world and from within themselves.

These interfaces form the entry layer of the NAW Human Interface Framework (NAW HIF), serving as the starting point for all perception, interpretation, and action.

The purpose of this domain is to:

- Identify the natural sensory and internal pathways that structure human experience
- Provide a stable reference for analyzing perception across contexts and devices
- Enable reproducible understanding of how information enters the human system
- Support preventive and durable design by grounding all interaction in natural human capacities

The six interfaces—Visual, Auditory, Tactile, Olfactory, Gustatory, and Internal Sense—constitute a complete and non-redundant set. Together, they describe the full range of human perceptual input.

2.2 Structure of the Primary Interface Domain

The Primary Interface Domain is defined by the following characteristics:

Natural Origin

Each interface arises from inherent human physiology and cognition, independent of technological or cultural conditions.

Non-redundancy

The six interfaces do not overlap in function; each provides a distinct perceptual channel.

Universality

These interfaces are shared across human populations and remain stable across historical periods.

Integration

Although distinct, the interfaces operate simultaneously and contribute to unified perception.

These characteristics ensure that the Primary Interface Domain functions as a durable foundation for all higher-level interaction.

2.3 The Six Primary Interfaces

2.3.1 Visual

Definition

Reception of information through sight, including shape, motion, color, and spatial patterns.

Role in the Framework

Visual input provides high-resolution spatial and structural information, supporting orientation, recognition, and prediction.

Key Properties

- High bandwidth
- Strong influence on attention
- Supports rapid pattern recognition

2.3.2 Auditory

Definition

Reception of information through sound, including tone, rhythm, and verbal or non-verbal cues.

Role in the Framework

Auditory input enables temporal tracking, communication, and environmental awareness beyond the visual field.

Key Properties

- 360-degree perception
- High temporal resolution
- Essential for social interaction

2.3.3 Tactile

Definition

Reception of information through physical contact, pressure, texture, and temperature.

Role in the Framework

Tactile input provides direct physical feedback, supporting manipulation, safety, and embodied learning.

Key Properties

- Immediate physical grounding
- Supports fine motor control
- Critical for object interaction

2.3.4 Olfactory

Definition

Reception of information through smell, enabling detection of environmental and chemical cues.

Role in the Framework

Olfactory input contributes to environmental assessment, memory formation, and internal state regulation.

Key Properties

- Strong emotional linkage
- Sensitive to environmental change

- Supports hazard detection

2.3.5 Gustatory

Definition

Reception of information through taste, supporting evaluation of substances and internal states.

Role in the Framework

Gustatory input provides chemical evaluation essential for safety, nutrition, and internal regulation.

Key Properties

- Direct chemical assessment
- Strong interaction with internal sense
- Supports survival-related decisions

2.3.6 Internal Sense

Definition

Perception of internal bodily and cognitive states, including balance, proprioception, tension, breathing, emotion, and mental orientation.

Role in the Framework

Internal Sense provides continuous feedback about the body's position, condition, and readiness, forming the basis for coordinated action and self-regulation.

Key Properties

- Continuous and background-dominant
- Integrates physical and cognitive states
- Essential for stability and orientation

2.4 Integration Across the Six Interfaces

Although each interface is defined independently, real-world perception emerges from their combined activity.

Key integration patterns include:

Multisensory Reinforcement

Visual and auditory cues often combine to strengthen recognition and prediction.

Embodied Feedback Loops

Tactile and Internal Sense work together to support movement, balance, and manipulation.

Environmental Assessment

Olfactory and Gustatory interfaces provide chemical and safety-related information that complements visual and auditory input.

Cross-Modal Compensation

When one interface is limited, others adapt to maintain functional perception.

These integration patterns form the perceptual foundation for the Contextual and Device domains described in later chapters.

2.5 Position Within the Three-Layer Model

The Primary Interface Domain is the first layer of the Three-Layer Model.

It defines how information enters the human system before it is shaped by context or expressed through devices.

Its position ensures:

- A stable starting point for all interaction
- A universal reference for analyzing perception
- A foundation for reproducible behavioral understanding
- A natural anchor for cross-domain dynamics

This layer is essential for interpreting the Contextual Interface Domain (Chapter 3) and the Interaction Device Domain (Chapter 4).

2.6 Summary

Chapter 2 defines the six natural interfaces that form the perceptual foundation of NAW HIF.

These interfaces—Visual, Auditory, Tactile, Olfactory, Gustatory, and Internal Sense—constitute a complete and universal set of human perceptual channels.

By establishing clear definitions and integration patterns, this chapter provides the structural basis for understanding how information enters the human system and how it interacts with context and devices in subsequent chapters.

Chapter 3. Contextual Interface Domain

3.1 Purpose of the Contextual Interface Domain

The Contextual Interface Domain defines the four situational modes that determine how information is received, processed, and integrated within the human system.

While the Primary Interface Domain describes the channels of perception, the Contextual Interface Domain describes the conditions under which perception occurs.

The purpose of this domain is to:

- Clarify how attention, environment, and cognitive state shape perception
- Provide a structural reference for analyzing behavior across varying levels of engagement
- Enable reproducible understanding of situational differences in learning and interaction
- Support preventive and durable design by accounting for natural contextual variation

The four contexts—Foreground, Background, Passive, and Ambient—form a complete and non-overlapping set that captures the full range of human perceptual conditions.

3.2 Structure of the Contextual Interface Domain

The Contextual Interface Domain is defined by the following characteristics:

Attentional Orientation

Each context reflects a distinct relationship between attention and incoming information.

Environmental Positioning

Contexts describe how the individual is situated relative to the surrounding environment.

Cognitive Load Distribution

Each context corresponds to a different allocation of mental resources.

Universality and Stability

These contexts appear across cultures, environments, and historical periods, making them durable structural units.

Together, these characteristics ensure that the Contextual Interface Domain provides a stable and universal framework for understanding situational perception.

3.3 The Four Contextual Interfaces

3.3.1 Foreground Interaction

Definition

Active, intentional engagement with a task or stimulus.

Role in the Framework

Foreground Interaction represents focused attention and deliberate processing.

It is the primary mode for learning, problem-solving, and precise action.

Key Properties

- High attentional investment
- Strong goal orientation
- High cognitive load

3.3.2 Background Interaction

Definition

Parallel or peripheral engagement occurring while attention is directed elsewhere.

Role in the Framework

Background Interaction supports multitasking, environmental monitoring, and the maintenance of ongoing activities without full attention.

Key Properties

- Peripheral awareness

- Moderate cognitive load
- Supports continuity of behavior

3.3.3 Passive Interaction

Definition

Reception of information without deliberate attention or effort.

Role in the Framework

Passive Interaction allows the human system to absorb information with minimal cognitive investment, supporting memory formation and environmental adaptation.

Key Properties

- Low attentional demand
- Often subconscious
- Supports incidental learning

3.3.4 Ambient Interaction

Definition

Continuous perception of environmental information as part of the surrounding context.

Role in the Framework

Ambient Interaction provides a stable sense of presence, safety, and orientation.

It forms the perceptual background against which all other interactions occur.

Key Properties

- Continuous and low-effort
- Strongly tied to Internal Sense
- Essential for spatial and emotional stability

3.4 Interaction Patterns Across Contexts

Although each context is defined independently, real-world behavior emerges from their dynamic interplay.

Key patterns include:

Context Cycling

Individuals naturally shift between contexts as tasks and environments change.

Foreground–Background Coordination

Focused tasks often rely on background processes for stability and continuity.

Passive Reinforcement

Information encountered passively can later support foreground learning.

Ambient Anchoring

Ambient cues provide the baseline orientation necessary for all other contexts.

These patterns illustrate how context shapes perception and action across time.

3.5 Relationship to the Primary Interface Domain

The Contextual Interface Domain determines how the six primary interfaces are utilized.

Examples:

- Visual input in Foreground mode supports detailed analysis
- Auditory input in Background mode supports environmental monitoring
- Internal Sense in Ambient mode maintains stability and orientation
- Tactile input in Passive mode supports incidental learning through contact

Thus, context modulates the intensity, priority, and interpretation of primary sensory input.

3.6 Position Within the Three-Layer Model

The Contextual Interface Domain is the second layer of the Three-Layer Model.

It shapes the conditions under which primary sensory information is processed and prepares the system for action through devices.

Its position ensures:

- A structural bridge between perception and action
- A universal reference for analyzing situational behavior
- A foundation for cross-domain dynamics
- A stable framework for understanding cognitive load and attention

This layer is essential for interpreting the Interaction Device Domain (Chapter 4).

3.7 Summary

Chapter 3 defines the four contextual modes that determine how information is received and processed within the human system.

Foreground, Background, Passive, and Ambient interactions form a complete and universal set of perceptual contexts.

By establishing clear definitions and interaction patterns, this chapter provides the structural basis for understanding how

situational factors shape perception and how they interact with sensory input and device-based action in subsequent chapters.

Chapter 4. Interaction Device Domain

4.1 Purpose of the Interaction Device Domain

The Interaction Device Domain defines the two fundamental means through which humans act upon, manipulate, or communicate with the external world.

While the Primary Interface Domain describes how information enters the human system, and the Contextual Interface Domain describes under what conditions it is processed, the Interaction Device Domain describes how humans express action.

The purpose of this domain is to:

- Identify the natural and mediated pathways through which human intention becomes external action
- Provide a structural reference for analyzing interaction across physical and digital environments
- Enable reproducible understanding of how behavior is executed
- Support preventive and durable design by grounding all interaction in natural human capacities

The two devices—Direct Device and Mediated Device—form a complete and non-redundant set that captures all forms of human action.

4.2 Structure of the Interaction Device Domain

The Interaction Device Domain is defined by the following characteristics:

Action Orientation

Each device represents a distinct pathway through which intention is transformed into action.

Physical vs. Mediated Contact

The domain distinguishes between direct bodily engagement and technologically mediated engagement.

Universality

Both devices appear across cultures and historical periods, making them stable structural units.

Scalability

The domain accommodates both traditional tools and emerging digital systems without modification.

These characteristics ensure that the Interaction Device Domain remains durable and future-proof.

4.3 The Two Interaction Devices

4.3.1 Direct Device

Definition

A tool or medium operated through direct physical or cognitive contact (e.g., hands, voice, paper, body movement).

Role in the Framework

Direct Devices represent the most immediate form of action.

They rely on the body's natural capacities and require minimal mediation.

Key Properties

- High immediacy
- Strong embodiment
- Low technological dependency
- Supports fine motor control and expressive action

Examples

Handwriting, speaking, gesturing, manipulating objects, drawing, physical navigation.

4.3.2 Mediated Device

Definition

A tool or medium requiring technological or physical mediation (e.g., screens, controllers, digital interfaces).

Role in the Framework

Mediated Devices extend human capability beyond natural physical limits, enabling interaction with digital, remote, or abstract environments.

Key Properties

- High scalability
- Variable latency
- Dependent on external systems
- Enables complex or remote interaction

Examples

Smartphones, computers, VR systems, remote controls, digital dashboards.

4.4 Interaction Patterns Across Devices

Although Direct and Mediated Devices are defined independently, real-world action emerges from their combined use.

Key patterns include:

Sequential Device Use

Actions often begin with direct engagement and transition into mediated interaction (e.g., writing notes by hand before entering them digitally).

Parallel Device Use

Direct and mediated actions may occur simultaneously (e.g., speaking while using a touchscreen).

Device Switching

Users shift between devices depending on context, cognitive load, or task requirements.

Embodied–Digital Integration

Direct bodily movement often anchors mediated digital interaction, enhancing stability and orientation.

These patterns illustrate how devices shape the execution of human intention.

4.5 Relationship to the Primary and Contextual Domains

The Interaction Device Domain determines how perception and context are expressed as action.

Examples:

- Foreground + Direct Device → precise manual tasks
- Background + Mediated Device → passive digital monitoring
- Ambient + Direct Device → natural movement and navigation
- Passive + Mediated Device → low-effort digital consumption

Thus, the device domain completes the cycle of perception → context → action.

4.6 Position Within the Three-Layer Model

The Interaction Device Domain is the third layer of the Three-Layer Model.

It defines the pathways through which processed information becomes external behavior.

Its position ensures:

- A clear endpoint for the perceptual–cognitive–behavioral cycle
- A universal reference for analyzing action across environments
- A foundation for cross-domain dynamics
- A stable structure for understanding human–technology interaction

This layer completes the structural model introduced in Chapter 1.

4.7 Summary

Chapter 4 defines the two fundamental devices through which humans act upon the world.

Direct Devices and Mediated Devices form a complete and universal set of action pathways.

By establishing clear definitions and interaction patterns, this chapter provides the structural basis for understanding how human intention becomes behavior and how device-based action interacts with sensory input and contextual conditions.

Chapter 5. Cross-Domain Dynamics

5.1 Purpose of Cross-Domain Dynamics

Cross-Domain Dynamics describes the integrated behavior that emerges when the Primary Interface Domain, the Contextual Interface Domain, and the Interaction Device Domain operate together as a unified system.

While Chapters 2–4 define each domain independently, real-world human activity arises from their continuous interaction.

The purpose of this chapter is to:

- Explain how perception, context, and action interact dynamically
- Provide a structural model for analyzing real-time human behavior
- Enable reproducible understanding of transitions across layers and domains
- Support preventive and durable design by clarifying how interactions evolve over time

Cross-Domain Dynamics forms the operational core of NAW HIF, connecting static definitions to real-world behavior.

5.2 Nature of Cross-Domain Dynamics

Cross-Domain Dynamics is defined by the following characteristics:

Simultaneity

Multiple domains operate at the same time, even when one appears dominant.

Transition

Human activity naturally shifts between domains as tasks, environments, and internal states change.

Integration

Perception, context, and device use combine to form coherent behavior.

Stability and Variability

The system maintains overall stability while allowing flexible transitions.

These characteristics ensure that NAW HIF can describe both predictable patterns and adaptive behavior.

5.3 Core Components of Cross-Domain Dynamics

5.3.1 Interaction Flow

Definition

The movement of information across multiple domains during perception or action.

Function

Interaction Flow describes how sensory input, contextual conditions, and device-based actions influence one another in real time.

Examples

- Visual input → Foreground → Direct Device
- Auditory input → Background → Mediated Device

5.3.2 Domain Transition

Definition

Shifts between interface domains as perception or behavior changes.

Function

Domain Transition explains how individuals move from one domain to another, such as shifting from passive listening to active engagement.

Examples

- Passive → Foreground
- Direct Device → Mediated Device

5.3.3 Context Shift

Definition

Changes in situational or attentional context.

Function

Context Shift captures how attention reallocates in response to internal or external triggers.

Examples

- Background → Foreground when a task becomes urgent
- Foreground → Ambient during rest or recovery

5.3.4 Processing Pathway

Definition

The route through which information travels across layers and domains.

Function

Processing Pathways describe the structural sequence of perception → interpretation → action.

Examples

Visual → Foreground → Direct Device

Internal Sense → Ambient → Mediated Device

5.3.5 Interaction State

Definition

A temporary configuration indicating which domain, layer, and context are currently active.

Function

Interaction States provide a snapshot of the system at a given moment, enabling precise analysis of behavior.

Examples

- “Visual + Foreground + Direct Device”
- “Auditory + Background + Mediated Device”

5.4 Dynamic Patterns Across Domains

Cross-Domain Dynamics produces recognizable patterns that appear across tasks and environments.

5.4.1 Perception-Driven Patterns

Sensory input initiates shifts in context or device use.

Example: A sudden sound (Auditory) triggers a shift from Background to Foreground.

5.4.2 Context-Driven Patterns

Changes in attention or environment reshape perception and action.

Example: Entering a quiet space shifts interaction toward Ambient.

5.4.3 Device-Driven Patterns

The choice of device influences perception and context.

Example: Using a smartphone (Mediated Device) increases reliance on Visual and reduces Tactile engagement.

5.4.4 Stabilization Patterns

Internal Sense and Ambient Interaction provide baseline stability for all other dynamics.

These patterns allow NAW HIF to describe both micro-level and macro-level behavior.

5.5 Cross-Domain Dynamics in Real-World Scenarios

Cross-Domain Dynamics can be observed in everyday activities:

Example 1: Learning

Visual → Foreground → Direct Device

Auditory → Background → Mediated Device

Example 2: Navigation

Internal Sense → Ambient → Direct Device

Visual → Foreground → Direct Device

Example 3: Digital Interaction

Visual → Foreground → Mediated Device

Tactile → Background → Mediated Device

These examples illustrate how domains combine to form coherent behavior.

5.6 Relationship to the Three-Layer Model

Cross-Domain Dynamics is the operational expression of the Three-Layer Model.

It describes how:

- Primary interfaces supply perceptual input
- Contextual interfaces shape situational processing
- Device interfaces express action

This dynamic interaction completes the perceptual–cognitive–behavioral cycle defined in Chapter 1.

5.7 Summary

Chapter 5 defines the dynamic processes that connect perception, context, and action within NAW HIF.

Interaction Flow, Domain Transition, Context Shift, Processing Pathway, and Interaction State form the core components of Cross-Domain Dynamics.

By establishing these dynamic structures, this chapter provides the foundation for understanding real-time human behavior and

prepares the framework for the application examples presented in Chapter 6.

Chapter 6. Application Examples

6.1 Purpose of Application Examples

This chapter illustrates how the NAW Human Interface Framework (NAW HIF) can be applied to real-world activities. While Chapters 2–5 define the structural and dynamic components of the framework, this chapter demonstrates how those components combine to form practical, reproducible patterns of human behavior.

The purpose of this chapter is to:

- Show how the 6-4-2 structure appears in everyday tasks
- Provide clear examples of cross-domain integration
- Demonstrate how perception, context, and device use interact
- Offer a reference for applying NAW HIF in design, education, and analysis

These examples are not exhaustive; rather, they illustrate how NAW HIF can be used as a universal analytical tool.

6.2 Example 1: Learning and Study Behavior

Learning activities provide a clear demonstration of how sensory input, context, and device use interact.

Typical Configuration

- Primary Interface: Visual + Auditory
- Context: Foreground
- Device: Direct Device or Mediated Device

Processing Pathway

Visual → Foreground → Direct Device

Auditory → Background → Mediated Device

Explanation

- Visual input supports detailed reading and comprehension
- Auditory input provides supplementary cues (lectures, explanations)
- Foreground context enables focused learning
- Direct Devices (e.g., handwriting) reinforce memory
- Mediated Devices (e.g., digital materials) expand access

Key Insight

Learning is a multisensory, multi-context process that relies on stable Foreground engagement supported by Background and Passive cues.

6.3 Example 2: Physical Navigation

Navigation integrates sensory input, internal feedback, and direct bodily action.

Typical Configuration

- Primary Interface: Visual + Internal Sense
- Context: Ambient + Foreground
- Device: Direct Device

Processing Pathway

Internal Sense → Ambient → Direct Device

Visual → Foreground → Direct Device

Explanation

- Internal Sense provides balance, orientation, and spatial awareness
- Ambient context maintains environmental stability
- Visual input guides direction and obstacle avoidance
- Direct bodily movement expresses action

Key Insight

Navigation depends on the continuous integration of Internal Sense and Ambient context, forming a stable base for Foreground adjustments.

6.4 Example 3: Digital Interaction

Digital environments highlight the interaction between natural perception and mediated devices.

Typical Configuration

- Primary Interface: Visual + Tactile
- Context: Foreground + Background
- Device: Mediated Device

Processing Pathway

Visual → Foreground → Mediated Device

Tactile → Background → Mediated Device

Explanation

- Visual input dominates digital interaction
- Tactile input provides confirmation and control
(touchscreens, keyboards)
- Foreground context supports focused tasks
- Background context handles notifications and peripheral cues

Key Insight

Digital interaction is a hybrid process where mediated devices reshape the balance between sensory channels and contextual modes.

6.5 Example 4: Conversation and Social Interaction

Human communication integrates multiple sensory channels and dynamic context shifts.

Typical Configuration

- Primary Interface: Auditory + Visual + Internal Sense
- Context: Foreground ↔ Background
- Device: Direct Device

Processing Pathway

Auditory → Foreground → Direct Device

Visual → Background → Direct Device

Internal Sense → Ambient → Direct Device

Explanation

- Auditory input carries primary meaning
- Visual cues support interpretation (expressions, gestures)
- Internal Sense regulates emotional and physical presence
- Context shifts naturally between Foreground and

Background

Key Insight

Social interaction is inherently multi-layered, requiring rapid transitions across contexts and interfaces.

6.6 Example 5: Rest and Recovery

Rest is an essential application of NAW HIF because it demonstrates low-effort, stability-oriented dynamics.

Typical Configuration

- Primary Interface: Internal Sense
- Context: Ambient + Passive
- Device: Direct Device

Processing Pathway

Internal Sense → Ambient → Direct Device

Explanation

- Internal Sense monitors bodily state
- Ambient context maintains environmental stability
- Passive context allows low-effort perception
- Direct bodily posture expresses minimal action

Key Insight

Rest is a structured state defined by reduced sensory load and stable Ambient context.

6.7 Example 6: Creative Work

Creative tasks combine focused attention with flexible sensory integration.

Typical Configuration

- Primary Interface: Visual + Internal Sense + Tactile

- Context: Foreground + Background
- Device: Direct Device or Mediated Device

Processing Pathway

Visual → Foreground → Direct Device

Internal Sense → Background → Direct Device

Tactile → Foreground → Direct Device

Explanation

- Visual input guides composition and structure
- Internal Sense supports emotional and cognitive orientation
- Tactile input provides embodied feedback
- Context shifts support exploration and refinement

Key Insight

Creativity emerges from the interplay of Foreground focus and Background exploration.

6.8 Using NAW HIF as an Analytical Tool

These examples demonstrate how NAW HIF can be used to:

- Analyze behavior with structural clarity
- Identify perceptual and contextual bottlenecks
- Improve learning and design environments
- Support preventive and durable system design

- Provide a universal vocabulary for describing human activity

The framework's simplicity and universality allow it to be applied across disciplines, cultures, and technologies.

6.9 Summary

Chapter 6 illustrates how the NAW Human Interface Framework operates in real-world scenarios.

By applying the 6-4-2 structure to learning, navigation, digital interaction, communication, rest, and creative work, this chapter demonstrates the framework's versatility and explanatory power. These examples complete the NAW HIF Definition Document by showing how structural and dynamic components translate into practical understanding and design.

Afterword

The NAW Human Interface Framework was not built to explain the past.

It was built to endure the future.

In a world where technologies change faster than language, and attention fragments across devices and contexts, we need a structure that holds.

NAW HIF is that structure.

It does not depend on trends, platforms, or ideologies.

It depends only on the natural architecture of human perception, context, and action.

If this framework helps you see more clearly, design more responsibly, or teach more effectively, then it has fulfilled its purpose.

Thank you for reading.

Author's Note

This work is the result of recursive refinement.

Each term, each layer, each diagram was tested against the question:

“Will this still make sense in 100 years?”

I chose artificial terminology where necessary, avoided semantic drift, and prioritized clarity over popularity.

The goal was not to be accepted quickly, but to remain useful indefinitely.

I believe in frameworks that teach themselves.

If NAW HIF feels intuitive, it's because it reflects something already present in you.

My role was simply to reveal it.

Future Directions

NAW HIF is the first layer of a larger system.

It connects naturally to:

- Natural Brain I (NDB): Human intelligence as a Natural Data Base
- Natural Brain II (CIM): Comprehensive Information Management
- NAW Clubs: Educational environments for lifelong transferable skills

Future publications will expand these systems, always preserving the principles of minimalism, durability, and universal applicability.

The long-term goal is simple:

To build a complete architecture for human-machine coexistence that remains valid for a century.

About the Author

S. Hara is a system architect, educator, and framework designer.

He specializes in creating long-term durable structures for cognition, movement, and human interface.

His work blends historical vigilance with conceptual minimalism, aiming to transmit transferable skills across generations.

He prefers quiet influence over public exposure, and designs systems that teach themselves.

His frameworks are used in education, design, and strategic publishing worldwide.

This document was written in solitude, with clarity as its only audience.